

TIMMINS, ON, Canada

WMO: 717390

Lat: 48.568N

Lon: 81.378W

Elev: 967

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-21.7	-35.3	0.8	-27.4	-29.5	1.1	-21.3	20.6	8.9	18.8	6.9	3.7	280	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	85.1	67.7	81.6	65.4	78.5	64.0	70.5	80.6	68.4	77.7	66.5	74.9	9.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	103.4	75.1	65.2	96.4	73.2	63.1	89.7	71.0	35.1	81.0	33.3	77.8	31.7	75.1	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	17.2	14.7	-35.8	90.8	5.9	3.5	-40.1	93.3	-43.6	95.4	-46.9	97.3	-51.2	99.8		
			-36.0	74.3	5.9	1.9	-40.2	75.6	-43.7	76.7	-47.0	77.7	-51.3	79.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	3.0	6.9	19.2	34.0	49.5	59.7	63.9	61.4	53.8	40.9	26.8	12.4	(6)
(7)		DBStd	23.66	14.54	13.05	13.60	10.34	9.29	8.02	6.36	6.60	8.60	8.96	11.05	13.35	(7)
(8)		HDD50	6480	1457	1207	958	487	127	14	0	2	56	311	697	1164	(8)
(9)		HDD65	10707	1922	1627	1421	931	489	195	99	152	350	746	1146	1629	(9)
(10)		CDD50	1411	0	0	2	7	110	306	430	356	169	30	1	0	(10)
(11)		CDD65	162	0	0	0	0	7	37	64	41	13	0	0	0	(11)
(12)		CDH74	1916	0	0	2	5	142	518	705	409	131	4	0	0	(12)
(13)		CDH80	478	0	0	0	1	35	145	183	89	25	0	0	0	(13)
(14)	Wind	WSAvg	7.3	7.3	7.8	8.1	8.2	7.8	6.9	6.3	6.2	6.6	7.3	7.7	7.3	(14)
(15)	Precipitation	PrecAvg	36.3	2.4	1.9	2.6	2.1	3.0	3.8	3.9	3.8	3.7	3.0	3.2	2.7	(15)
(16)		PrecMax	44.5	5.7	4.1	6.0	4.3	5.6	8.7	7.6	8.1	6.9	6.4	7.0	4.8	(16)
(17)		PrecMin	29.8	0.7	0.5	0.8	0.3	0.9	1.1	1.8	1.0	2.2	1.5	0.8	1.1	(17)
(18)		PrecStd	3.9	1.0	0.9	1.2	0.8	1.2	1.6	1.4	1.6	1.1	1.1	1.4	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	43.4	59.3	70.7	85.1	89.5	89.8	87.1	83.3	72.6	57.3	40.4	(19)
(20)		MCWB	36.0	39.1	49.3	52.5	64.1	68.7	70.9	69.7	67.4	59.8	49.6	38.0	(20)	
(21)		2%	DB	33.5	36.5	49.5	62.1	77.6	84.1	84.8	82.2	77.6	66.0	50.7	36.6	(21)
(22)		MCWB	32.4	33.6	41.5	47.4	59.0	66.4	67.9	67.2	65.2	56.8	46.5	34.9	(22)	
(23)		5%	DB	30.6	31.4	43.5	56.2	72.7	80.0	81.1	78.7	73.2	60.6	45.7	33.9	(23)
(24)		MCWB	29.3	29.1	37.1	44.1	56.7	62.7	65.7	65.7	62.7	54.3	42.2	32.6	(24)	
(25)		10%	DB	26.1	27.0	38.5	50.4	67.8	75.9	77.8	74.9	68.8	55.5	41.4	31.5	(25)
(26)		MCWB	24.8	25.0	33.5	41.1	54.4	61.3	64.5	63.5	60.8	50.5	38.8	30.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	39.5	51.9	55.7	68.6	71.9	73.2	72.4	70.2	62.7	51.9	39.1	(27)
(28)		MCDWB	37.4	42.0	56.7	65.9	77.3	85.3	85.3	82.1	78.6	69.0	54.8	40.1	(28)	
(29)		2%	WB	32.5	33.3	42.4	48.9	62.9	68.9	70.6	69.4	67.0	58.4	47.3	35.1	(29)
(30)		MCDWB	33.2	36.4	47.9	58.7	73.4	79.8	80.4	78.7	74.6	64.2	50.2	36.2	(30)	
(31)		5%	WB	29.2	29.2	37.5	45.3	58.7	66.2	68.4	67.2	64.6	54.9	42.6	32.7	(31)
(32)		MCDWB	30.5	31.1	43.1	54.6	68.1	75.1	77.5	75.8	71.2	60.6	45.2	33.8	(32)	
(33)		10%	WB	24.7	25.2	33.8	41.7	55.4	63.5	66.3	65.1	61.8	50.8	38.8	30.3	(33)
(34)		MCDWB	26.2	27.0	38.0	49.9	66.2	71.9	75.0	73.0	68.0	55.4	41.5	31.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	19.5	21.8	22.6	21.0	23.9	23.6	22.4	21.5	20.3	15.4	13.0	15.2	(35)	
(36)		5% DB	MCDBR	17.1	19.5	24.5	30.7	32.4	30.0	27.2	26.6	25.7	23.2	17.6	13.6	(36)
(37)		MCWBR	16.4	17.3	17.4	18.6	17.4	14.8	13.2	13.7	14.8	15.4	14.0	12.9	(37)	
(38)		5% WB	MCDBR	16.6	18.8	21.5	27.4	26.5	23.8	23.0	22.5	22.2	20.9	16.4	13.0	(38)
(39)		MCWBR	16.1	17.1	16.0	18.2	16.3	13.7	12.5	12.8	13.9	15.3	13.9	12.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.274	0.271	0.299	0.341	0.358	0.377	0.391	0.375	0.360	0.323	0.287	0.258	(40)	
(41)		taud	2.207	2.246	2.257	2.323	2.417	2.370	2.363	2.405	2.433	2.521	2.483	2.331	(41)	
(42)		Ebn at Noon	247	281	291	288	286	280	274	274	267	262	249	244	(42)	
(43)		Edh at Noon	25	30	35	37	35	37	37	34	30	23	19	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	407	747	1206	1499	1656	1788	1720	1444	1079	631	337	266	(44)	
(45)		RadStd	66	99	116	131	136	114	152	86	84	84	34	40	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page